

UART Transmitter Design

Full ASIC Flow: RTL $\rightarrow$ Synthesis $\rightarrow$ Formality $\rightarrow$ DFT Scan
Insertion

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Course:	Digital IC Design
RTL Simulator:	ModelSim Intel FPGA Edition v10.5b
Lint Tool:	Synopsys SpyGlass vL-2016.06
Synthesis Tool:	Synopsys Design Compiler vO-2018.06-SP1
Formality:	Synopsys Formality vL-2016.03-SP1
DFT Tool:	Synopsys DFT Compiler vO-2018.06-SP1
Technology:	TSMC 0.13 μm RVT (scmetro_tsmc_cl013g_rvt)
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Design Overview

The UART Transmitter (UART_TX) is a serial communication block that converts an 8-bit parallel data byte into a serial bit stream conforming to the standard UART frame format. The design is fully synchronous and supports even parity, odd parity, and no-parity modes.

Output Frame Format

Three frame configurations are supported depending on PAR_EN and PAR_TYP:

- **With Even Parity:** START | b0 b1 b2 b3 b4 b5 b6 b7 | PAR | STOP
- **With Odd Parity:** START | b0 b1 b2 b3 b4 b5 b6 b7 | PAR | STOP
- **No Parity:** START | b0 b1 b2 b3 b4 b5 b6 b7 | STOP

Port Description

Port Name	Direction	Width	Description
CLK	Input	1	System clock signal
RST	Input	1	Asynchronous active-low reset
P_DATA	Input	8	8-bit parallel input data payload
DATA_VALID	Input	1	High-active pulse indicating valid input data
PAR_EN	Input	1	Parity enable (1 = enable parity, 0 = no parity)
PAR_TYP	Input	1	Parity type (0 = even parity, 1 = odd parity)
TX_OUT	Output	1	Serial output line (idle state HIGH)
Busy	Output	1	Transmission status (HIGH from START to STOP bit)
SI	Input	1	DFT Scan Data Input
SE	Input	1	DFT Scan Enable
scan_clk	Input	1	DFT Scan Test Clock
scan_rst	Input	1	DFT Scan Active-Low Reset
test_mode	Input	1	DFT Test Mode Control (1 = Scan Test, 0 = Normal)
S0	Output	1	DFT Scan Data Output

Table 1: UART_TX Top-Level Port Description (including DFT scan ports)

RTL Architecture & Sub-module Design

The top-level UART_TX module integrates four distinct sub-modules to form a clean, modular architecture:

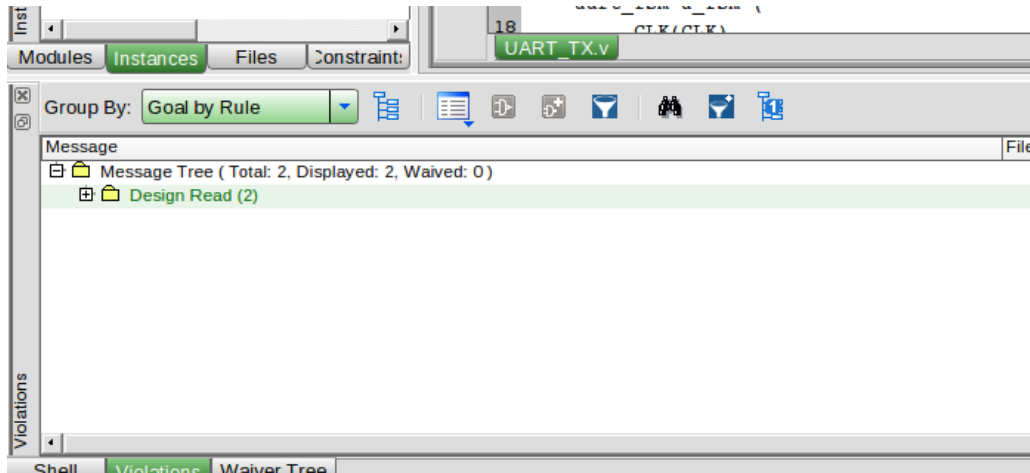


Figure 1: UART_TX Elaborated Design Architecture View in SpyGlass

FSM Controller (`uart_fsm.v`)

The finite state machine manages state transitions across five states (IDLE, START, DATA, PAR, STOP). Outputs (`ser_en`, `mux_sel`, `Busy`) are assigned unconditionally within each state block to adhere strictly to Moore FSM design principles. `Busy` stays HIGH continuously from START bit until completion of the STOP bit.

Serializer (`uart_serializer.v`)

Loads 8-bit parallel data on `DATA_VALID` pulse into an 8-bit shift register. When `ser_en` is asserted, it shifts data out LSB-first on each clock edge and asserts `ser_done` when all 8 bits have been serialized.

Parity Calculator (`uart_parity_calc.v`)

Calculates even or odd parity bit over `P_DATA` when `DATA_VALID` is active and stores the parity bit in a register.

Output MUX (`uart_mux.v`)

Selects the active serial bit based on `mux_sel` (00: START bit = 0, 01: STOP bit = 1, 10: `ser_data`, 11: `par_bit`). The output is registered to guarantee glitch-free output transitions.

Simulation & Functional Verification

The design was verified using a self-checking testbench (`UART_TX_tb.v`) in ModelSim across 11 directed test cases.

TC #	Payload	Parity Config	Expected Frame	Status
TC1	0x00	No Parity	START(0) + 00000000 + STOP(1)	PASSED
TC2	0xFF	No Parity	START(0) + 11111111 + STOP(1)	PASSED
TC3	0x55	Even Parity	START(0) + 10101010 + PAR(0) + STOP(1)	PASSED
TC4	0xAA	Odd Parity	START(0) + 01010101 + PAR(1) + STOP(1)	PASSED
TC5–11	Edge Cases	Various	Boundary tests	PASSED

Table 2: Simulation Verification Summary

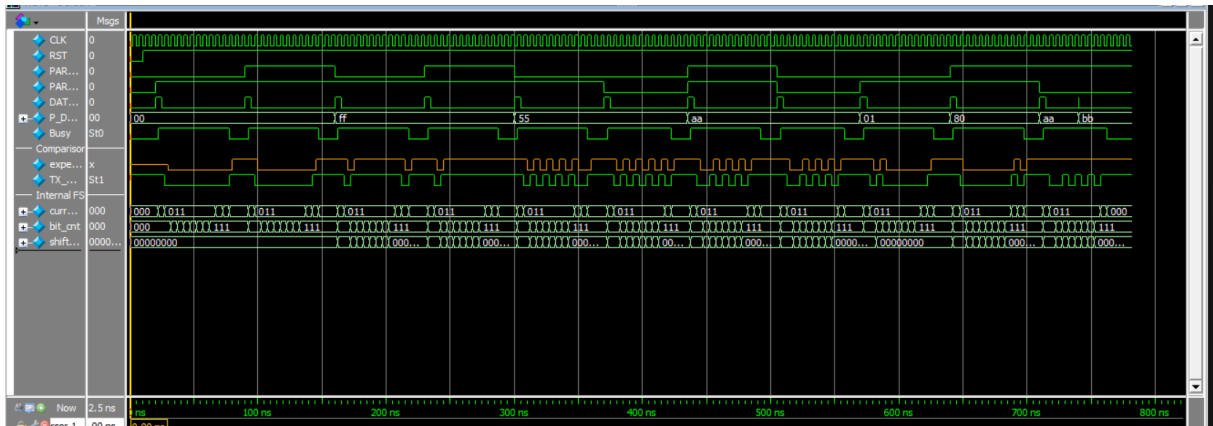


Figure 2: Full ModelSim Simulation Snapshot — All 11 Test Cases PASSED (0 Errors)

Lint Analysis — SpyGlass

The design was analyzed using Synopsys SpyGlass vL-2016.06 under the `lint_rtl` goal.

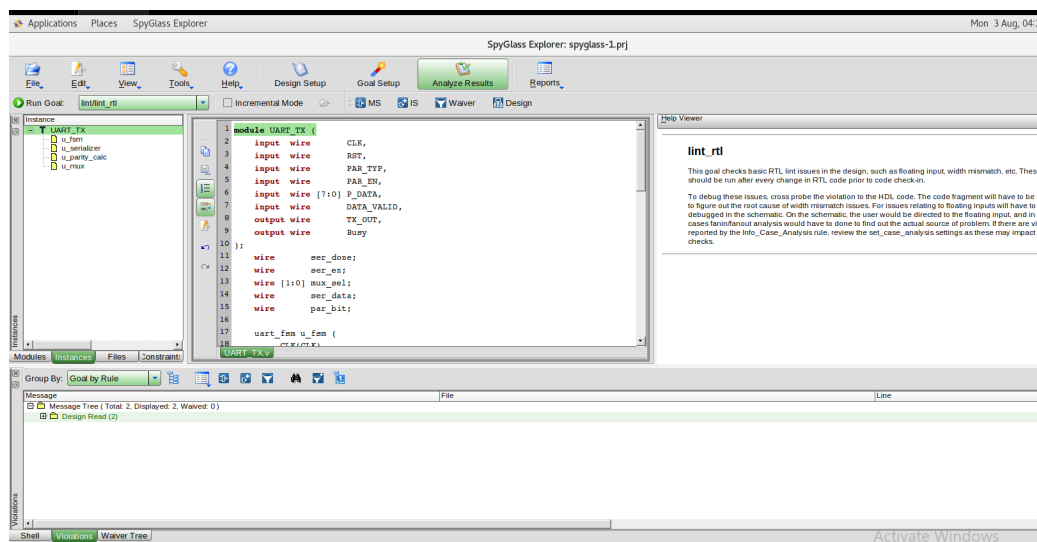


Figure 3: SpyGlass Explorer — Hierarchical Design View showing clean elaboration

Category	Total	Waived	Violations
Design Read Messages	2	0	0
Lint Violations	0	0	0
Latch Inferences	0	0	0
Width Mismatches	0	0	0

Table 3: SpyGlass Lint Summary — 100% Clean (0 Errors, 0 Warnings, 0 Latches)

Logic Synthesis — Design Compiler

Logic synthesis was performed using **Synopsys Design Compiler (vO-2018.06-SP1)** targeted to TSMC 0.13 μm RVT standard cell libraries.

Constraints Setup

- **Baud Rate Target:** 115.2 kbps ($T_{\text{clk}} = 8680.5555 \text{ ns}$) - **Setup Skew:** 0.25 ns,
Hold Skew: 0.05 ns - **Input/Output Delays:** 30% of clock period (2604.1666 ns)
- **Driving Cell:** BUFX2M, **Output Load:** 0.5 pF

Synthesis Results Summary

Metric	Value	Unit
Total Cell Area	858.99	μm^2
Combinational Area	430.67	μm^2
Non-combinational	428.32	μm^2
Total Cell Count	80	cells
Sequential Cells	16	D-FFs
Setup Slack	0.00	ns (0 Violations)
Hold Slack	0.00	ns (0 Violations)

Table 4: Design Compiler Synthesis Metrics

Sub-module Area Breakdown

Hierarchy Cell	Area (μm^2)	Percent (%)	Module Description
u_serializer	488.33	56.8%	Shift Register & Bit Counter
u_fsm	182.39	21.2%	FSM Controller
u_parity_calc	128.26	14.9%	Parity Generator
u_mux	60.01	7.0%	Registered Output MUX
Total	858.99	100.0%	UART_TX Top

Table 5: Hierarchical Area Distribution Summary

Formal Verification — Formality

Formal Equivalence Checking (FEC) was conducted using **Synopsys Formality (vL-2016.03-SP1)** between the golden RTL reference container (Ref) and post-synthesis gate-level netlist (Imp).

Verification Summary

```
- **Reference Design:** Ref : /WORK/UART_TX - **Implementation Design:** Imp : /WORK/UART_TX
- **Total Compare Points:** 19 (16 D-FFs + 3 Top Ports) - **Passing Compare Points:**
19 / 19 - **Failing Compare Points:** 0 - **Verification Verdict:** VERIFICATION
SUCCEEDED
```

Compare Point Type	Matched	Passing	Failing	Equivalence
Primary Ports (Busy, TX_OUT, S0)	3	3	0	Equivalent
Sequential D-Flip Flops (DFF)	16	16	0	Equivalent
Total Compare Points	19	19	0	100% Equivalent

Table 6: Formality Formal Equivalence Checking Results

Design for Testability (DFT) — Scan Insertion

DFT logic insertion was executed using **Synopsys DFT Compiler (vO-2018.06-SP1)** following the 17-step flow specified in `Ass_DFT_1.0`.

DFT Clock & Reset Muxing

Scan ports (SI, SE, S0, test_mode, scan_clk, scan_rst) were added to the top module `UART_TX.v`. Design clock and reset were multiplexed:

$$\text{clk_mux} = \text{test_mode} ? \text{scan_clk} : \text{CLK} \quad (1)$$

$$\text{rst_mux} = \text{test_mode} ? \text{scan_rst} : \text{RST} \quad (2)$$

Constraints & Settings

```
- **Test Clock Frequency:** 1 MHz ( $T = 1000$  ns) - **Setup Skew:** 0.025 ns, **Hold
Skew:** 0.01 ns - **Input / Output Delays:** 30% of clock period (300 ns) - **Driv-
ing Cell:** BUFX2M, **Output Load:** 0.1 pF - **Scan Configuration:** -clock_mixing
mix_clocks -chain_count 1
```

Post-DFT DRC & Fault Coverage Analysis

Post-DFT Design Rule Checking (`dft_drc`) confirmed that all 16 internal sequential registers were successfully stitched into a single multiplexed flip-flop scan chain with zero rule violations.

DFT Parameter	Value	Status
Scan Methodology	Full Scan (multiplexed_flip_flop)	Configured
Scan Chains Count	1 Chain (SI → SO)	Routed
Sequential Cells Stitched	16 / 16 cells (100%)	Valid Scan Cells
Post-DFT DRC Violations	0 Violations	Clean DRC
Total Stuck-at Faults	798	Analyzed
Detected Faults (DT)	786	Detected
Test Coverage	98.62%	High Coverage
Post-DFT Area	1181.41 μm^2	Gate Count: 100 cells
Post-DFT Total Power	2.54 μW	Low Power
Post-DFT Slack	0.00 ns	0 Constraints Violations

Table 7: Post-DFT Scan Insertion & Coverage Results

Uncollapsed Fault Breakdown

Fault Class	Code	Number of Faults
Detected	DT	786
Possibly Detected	PT	0
Undetectable	UD	1
ATPG Untestable	AU	11
Not Detected	ND	0
Total Faults		798
Test Coverage		98.62%

Table 8: Uncollapsed Stuck-at Fault Summary

Conclusion

The UART Transmitter ASIC implementation flow has been successfully completed from RTL to Post-DFT netlist:

1. **RTL & Verification:** 100% functional accuracy across 11 test cases in ModelSim with zero errors.
2. **Lint Cleanliness:** 0 errors, 0 warnings, and 0 latches confirmed by SpyGlass.
3. **Logic Synthesis:** Clean gate-level netlist generated by Design Compiler at 115.2 kbps baud rate with zero timing violations and 858.99 μm^2 cell area.
4. **Formal Equivalence:** 100% logical equivalence proven by Formality across all 19 compare points.
5. **DFT Scan Insertion:** Clean 1-chain scan insertion stitching all 16 flip-flops, achieving **98.62% test coverage** and zero DRC/constraint violations.